Note to grader:

I wanted my project to be more-or-less self-contained, so I'm going to document the technical details and A.I. usage log here, rather than on the site itself. I also included the literature review I wrote before beginning the project here, as it was too long and dependent on citations to fit neatly into the website.

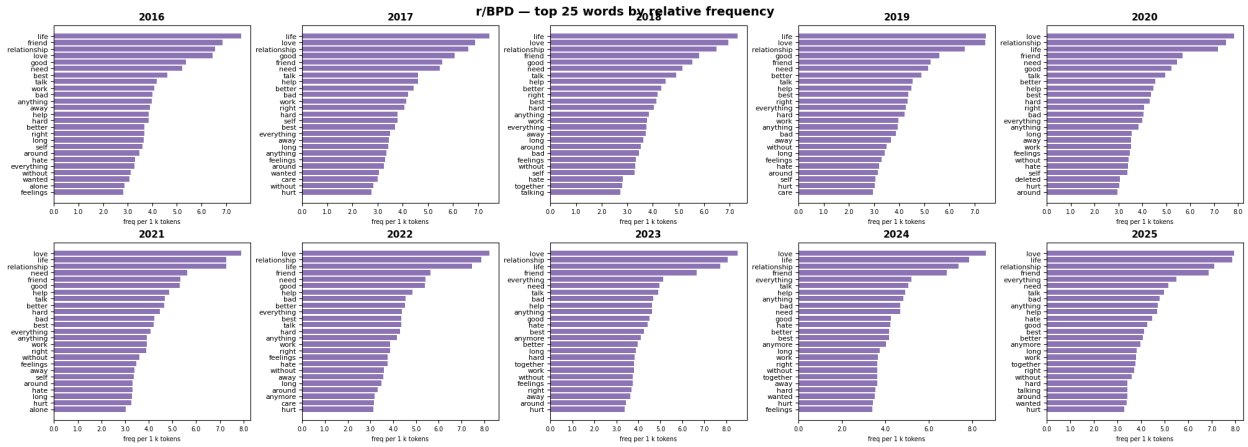
Technical appendix:

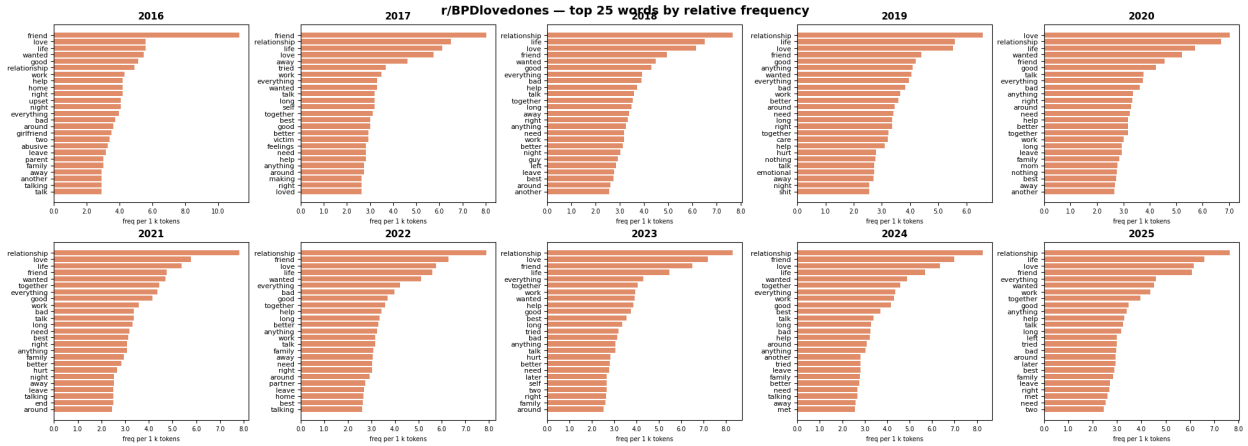
Data collection:

The Reddit data from r/BPD and r/BPDlovedones was loaded using an API called "Arctic Shift", which is essentially a massive pre-made archive of Reddit data. This allowed me to download massive amounts of data while also avoiding any web scraping restrictions, since there was no automated web crawling involved. After the posts were collected, I filtered out any duplicate posts or empty rows.

Word frequency analysis:

Before any language processing could begin, I filtered out any low-signal filler words. I used the default NLTK stopwords, but also added in some custom stopwords as I ran the frequency analysis. My intention was to find emotionally-valenced words that changed in frequency over time, and many of the top words (e.g. favorite, personality, disorder) ended up overrepresented and hid some of the more interesting words (e.g., hate). I saved the top 200 words for each year/subreddit pair. Here are some static graphs of the word frequency counts immediately after the WFA:





Sentiment and emotional analysis:

I ran the sentiment analysis using the GoEmotions library. I found a library which assigns 27 emotions to a chunk of text, with magnitudes that describe how strongly each emotion comes through. Since I was working with around 80,000 posts, I downsampled and analyzed only around 14,000, which was smaller, but still a good sample and feasible on my machine. I then determined the top sentiment per post, and aggregated the frequency of the top emotions in the dataset by year (which is what ended up on the final visualizations). I then grouped the 27 emotions into “positive” or “negative” categories to visualize more general trends in sentiment over time.

Co-occurrence and network diagram:

Each word in the corpus was tokenized, and then for each token, word pairings were created based on the next 5 words in the corpus. So for a sentence like “I love my favorite person”, the script would store 4 tuples for “I”: (“I”, “love”), (“I”, “my”), (“I”, “favorite”), (“I”, “person”). Once these pairs were created, the script filters for just the most commonly occurring pairs by count. The top pairings were then represented in a D3 force-directed network, with the edge thickness indicating the frequency of a given pair.

A.I. use log:

I used assistance from a Claude agent for most stages of the technical development of this project. For example, the A.I. handled the reformatting of the data from the Jupyter notebook so that it would be compatible with the D3 visualizations. I also used it to discuss the best ways to practically approach the analysis, such as which libraries to use for the data collection and NLP. However, much of the final presentation was very deliberately tweaked by hand, such as the look and feel of the css elements, and the way objects materialize and fade out as you scroll through the project. I was inspired by the visual essays on the website “pudding.cool”, and I think capturing that kind of flow and style wasn’t achievable purely through a single prompt. ***The literature review, interpretation of the charts, and the illustrations were all completed without any A.I. assistance. Anything written in plain English was my work, and not that of A.I.***

I had a lot of fun making this project, and plan to make similar projects in the future. My intention is to become less reliant on A.I. over time, and more capable of doing things like this by hand. Having to manually bugfix the html integration and D3 charts did significantly advance

my skills. However, this project would absolutely not have been possible without A.I., given my current knowledge of data visualization.

Literature review:

Introduction

Individuals with borderline personality disorder (BPD) commonly enter into intense, unstable close relationships, marked by conflict, anxiety, and rapid shifts between the idealization of a partner, to devaluation (American Psychological Association, 2013). In recent years, borderline individuals lacking close relationships in "offline" spaces have turned towards social media and internet message boards to seek community and emotional support. As of 2020, borderline users of all age groups report posting on social media more frequently than healthy users, and consider online activity more integral to their daily routines than non-borderlines (Ooi et al., 2020). Specialized online forums, such as r/BPD on reddit.com, allow borderlines and their loved ones to share their unique experiences, and build a supportive community. Like other niche online spaces, these forums have developed their own cultures, conventions, and language over time. One term popularized in recent years is the borderline "favorite person", or "FP".

The "FP" began circulation in online spaces such as Tumblr and Reddit, and eventually caught on within the wider borderline community. The first recorded instance of the term "favorite person" (to the author's knowledge) appears on a Tumblr blog from late 2015, titled "Shit Borderlines Do". While no standardized criteria define the FP relationship, the favorite person broadly refers to a friend or romantic partner with whom the borderline person is deeply and maladaptively infatuated. The borderline person relies upon the FP for constant attention,

affection, and reassurance, and experiences excruciating emotional distress when the FP cannot reciprocate their intensity. One user describes their FP relationship as follows:

"I'm like a dog who destroys the house when she's left at home, but then acts really happy when the owner comes home and pretends nothing happened… I will text my FP every morning when I wake up saying "good morning xx," and if they haven't replied within five minutes I automatically assume they either hate me or I have annoyed them. That thought sends me into utter panic and causes a lot of distress. If I were to think about it logically, I would probably tell myself they're probably still asleep — but when it comes to having an FP all rational thinking goes out of the window."

Search engine trends show a steady increase in interest in the FP over time, plateauing around late 2022. Clinicians and the general public have adopted the term, with online health magazines such as Verywell Mind and Medical News Today publishing digestible, educational articles on the borderline FP. The information presented in these articles, however, is mostly anecdotal, aggregated from counselor testimonials and individual self-reported forum posts. To-date, only two peer-reviewed, empirical reports examine the "FP" construct in detail.

A linguistic analysis of 207 text posts harvested from popular websites such as Reddit and Quora narrow down key features of the FP. The FP is a good listener, is sympathetic to the borderline person's troubles, and is someone with whom they feel safe and loved. Simultaneously, the FP is just as often associated with harm and painful emotions like anxiety, jealousy, and abandonment (Jeong et al., 2022). The borderline person evaluates situations as either "black or white", fluctuating between extreme idealization towards the FP, or disdain and avoidance. Borderlines often make self-injurious and suicidal gestures towards the FP, either to provoke a response, or to cope with the intensity of their emotions (Jeong et al., 2022). A

similar study of social media posts identified core features of the FP via factor analysis, which the authors labeled “self-confidence and identity disturbance” and “impaired mentalizing”. The borderline person reshapes their self-image to win the affection of the FP, modifying their interests, behavior, and appearance to suit the FP’s assumed preferences. They attempt to anticipate the FP’s internal feelings and intentions, even when their beliefs about the FP contradict reality (Stein & Johnson, 2025).

Profiling BPD in relation to the internet and digital cultures

One difficulty in BPD research relative to other cognitive disorders are the limited triggers which create social impairments. Unlike conditions such as autism and schizophrenia, characterized by social impairments in most or all social contexts, BPD symptoms tend to arise only under specific, momentary triggers, such as a perceived threat of abandonment (New & Joseph, 2018). Outside of situational triggers, emotional regulation and sociability hover around a baseline, meaning that people with BPD can often appear healthy and normative when examined in a laboratory (Miskewicz et al., 2015). This makes anecdotal data from the internet valuable, as online self-reports maintain some of the spontaneity of a borderline episode, free from technical and ethical restrictions. In fact, borderline internet users are more likely than healthy users to disclose intimate and personal information on the internet, making online data a useful supplement to the more rigorous data produced in a lab (Deutz et al., 2022).

Characteristics of borderline internet users

Though online data can be powerful, it may not be immediately generalizable to the wider population. Borderline users are twice as likely to meet the criteria for social media addiction than healthy users, and tend to use social media in maladaptive ways, such as the excessive monitoring of other’s social lives (Collins & Grant, 2025). Borderline individuals report

they frequently regret posting images and messages online, and are more likely than healthy individuals to delete or edit their posts (Ooi et al. 2020). Internet usage for borderline users correlates with higher levels of social isolation, which may worsen BPD symptoms and promote further dependence on internet use (Longshore, 2021). Finally, despite a predisposition to share personal information online, borderline users often selectively withhold information in order to curate a desirable, context-dependent self-image (Deutz et al., 2022).

Rationale for tracking linguistic changes in the borderline FP

Rates of BPD and borderline symptoms among internet-savvy young adults appear to have increased in recent years. A longitudinal study of undergraduate students at Southern Plains University found an upward trend in borderline symptoms and non-suicidal, self-injurious behavior between 2017 and 2021 (Min et al, 2023). This follows a broad increase in rates of depression, self-harm, and suicide among adolescents beginning in 2012, concurrent with rapid proliferation of smartphones and social media in the early 2010s (Twenge, 2020). Dependence on digital media is a key mechanism behind this observed change, as internet usage tends to exacerbate borderline symptoms (outlined above). However, another possible factor has yet to be examined: namely, shifts in the popular conceptualization of BPD as a whole.

Terminology in psychiatry is malleable, and the diagnostic criteria for a disorder often becomes more inclusive over time. This can occur through lowering the diagnostic threshold, or by classifying previously benign behaviors and cognitions as harmful and pathological (Haslam et al., 2021). Lay-adoption of psychiatric language is not without consequence. For example, after experiencing an unpleasant but harmless video clip, individuals with a broad definition of “trauma” were more likely to report feeling “traumatized” themselves, compared to

individuals with a narrow definition (Jones & McNally, 2022). Words used to describe undesirable situations become increasingly pathological and political; terms such as “trauma”, “abuse”, and “PTSD” magnify regrettable, yet common social infractions to the level of life-threatening violence (Haslam, 2021). Consequently, valuable attention and resources are diverted from the kinds of crises which the terms were first created to treat, and high quality mental healthcare becomes increasingly exclusive to those who need it the least. The concept of the FP is relevant in that it crystallizes an abstract concept, folk definitions of an FP developed in online fora, into a shorthand which implies a specific set of criteria and treatments. More specifically, it categorizes a certain kind of relationship, in its totality, as pathological. Within an individual’s lifetime, they will inevitably experience some form of “BPD” symptomology: identity disturbance, fear of abandonment, and/or excessive idealization and devaluation. However, given the nature of words to generalize beyond their initial conception, the “FP” might become common parlance, motivating individuals both with and without BPD to liberally psychiatrize their close relationships. The present study seeks to track changing conceptions of the borderline FP over time, identifying the ways in which it is dynamic, and the ways it is stable.

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